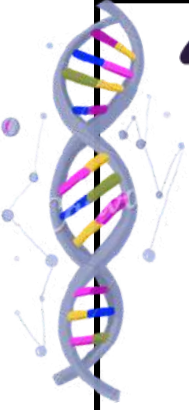
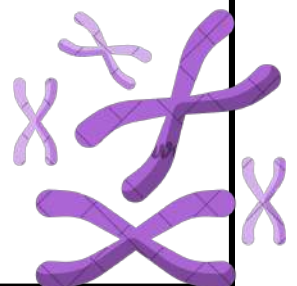


Canada



Final Exam



GeneticFinal Season

27) What is the underlying cause of dark urine, jaundice and low red blood cell count and hemoglobin concentration after intake of primaquine in a malaria patient?

- A. dihydropyrimidine dehydrogenase deficiency
- B. glucose-6-phosphate dehydrogenase deficiency
- C. cytochrome p450 deficiency
- D. thiopurine methyltransferase deficiency

28) Using genetic or genomic information to select the most appropriate choice of pharmacological therapy at the correct dosage for every person.

- A. Pharmacogenetics
- B. Pharmacogenomics
- C. personalized medicine
- D. Genetic variation

31) an increased in B-HCG with a reduced MSAFP suggests a possibility of which syndrome:

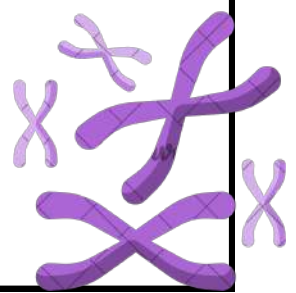
- A. Down syndrome
- B. Hydatidiform mole
- C. Neural tube defect

32) A 47-year-old female patient is referred to the genetic clinic, tumor in her right colo. Her father died at the age of 55 from colon cancer, her paternal aunt had endometrial cancer. What is the chance of her offspring to have colon cancer?

- A. %50
- B. %25
- C. %100
- D. %75

27	28	31	32
B	C	A	A

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GeneticFinal Season

34) Which of the following is considered a second line test for inborn errors of metabolism?

- A. Ammonia
- B. lactic acid
- C. blood gas
- D. urine organic acid

35) Which one of these conditions presents very similarly in Marfan's syndrome?

- A. Phenylketonuria
- B. Hurler disease
- C. Homocystinuria
- D. Tay-Sachs disease

36) Which of the following is related to lysosomal storage disorders?

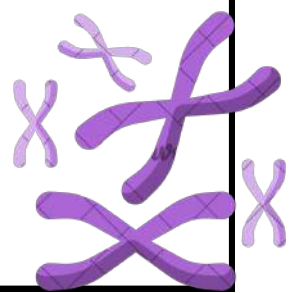
- A. Zellweger spectrum disorder
- B. Mucopolysaccharides

37) A 6 year old girl presented with intellectual disability, seizure, decreased pigmentation of skin, hair, iris and an odor smell of urine(mousy). Which amino acid would you expect to be elevated in the result of quantitative plasma amino acid profile?

- A. glycine
- B. homocysteine
- C. phenylalanine
- D. tyrosine

34	35	36	37
D	C	B	C

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GeneticFinal Season

41) A female with short stature and rhizomelic shortening of the bone. If the test showed Heterozygous mutation. What is the percentage to have an unaffected child if she's married to a man with the same gene mutation?

- A. 100%
- B. 1-3%
- C. 25%
- D. 50%

48) (case) Lentivirus therapy used in ..

- A. Severe anemia
- B. Neurodegenerative disease
- C. Myopathy disease

49) What is the meaning of consultand?

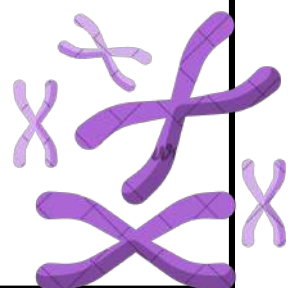
- A. an individual who seeks genetic counseling
- B. Who performs the psychological support

50) Which of the following best describes genetic counseling?

- A. It Focuses only on medical treatment of the patient
- B. It is done only by a clinical Geneticist
- C. it focuses only on psychological aspect of patient
- D. It deals with the risk of developing or transmitting a genetic disorder

41	48	49	50
C	A	A	D

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GeneticFinal Season

51) Which of the following is an indication for genetic counseling in adults?

- A. Family history of Appechiertis
- B. Adult with migraine headaches
- C. Family history of heart disease after 60 yea of age
- D. A strong personal or family history of cancer with heritability

52) Which of the following is a skill of genetic counseling?

- A. should be patient centered
- B. should be directive
- C. should be judgemental
- D. should be one-way process

54) What mutation causes malignant hyperthermia?

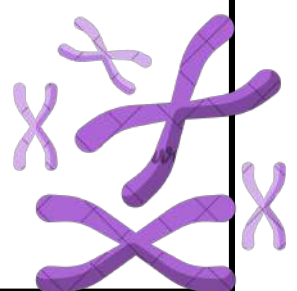
- A. Mutation in RYR1
- B. Mutation in pCHE
- C. Mutation in TPMT
- D. Mutation in DPYD

55) A patient experienced prolonged apnea after being given succinylcholine, what enzyme do they have a deficiency in?

- A. pseudocholinesterase
- B. Dihydropyrimidine Dehydrogenase
- C. Thiopurine methyltransferase

51	52	54	55
D	A	A	A

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GeneticFinal Season

61) Fast acetylators of isoniazid medication of tuberculosis usually suffer from?

- A. high failure of isoniazid in treating tuberculosis
- B. drug induced systemic lupus erythematosus
- C. increase risk of bladder cancer

62) The mutated gene in neonatal diabetes is:

- A. KCNJ11
- B. TPMT

65) An adult hemoglobin molecule consists of:

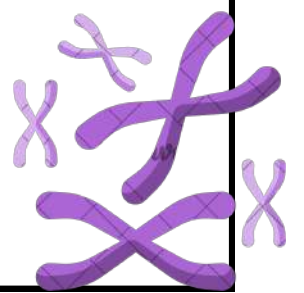
- A. 4 heme groups with 2 alpha chains and 2 beta chains
- B. 4 heme groups with 2 alpha chains and 1 gamma chain
- C. 3 heme chains with 2 alpha chains and 2 beta chains
- D. 3 heme chains with 2 alpha chains and 1 gamma chain

66) In Ornithine transcarbamylase deficiency, what is the investigation to be seen?

- A. Reduced orotic acid
- B. Lactic acid
- C. Hyperammonemia
- D. Increased methionine and homocysteine

61	62	65	66
A	A	A	C

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GeneticFinal Season

68) Which will occur if there is a loss of 4 α -globins?

- A. Cooley's Anemia
- B. δ beta thalassemia
- C. HbH
- D. Hydrops fetalis

71) A molecular analysis shows a mutation of the triplet codon for glutamic acid. Which of the following will be affected?

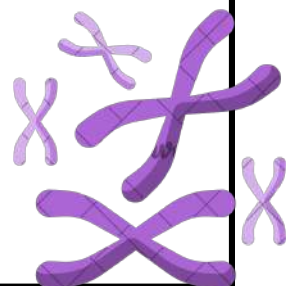
- A. HbS
- B. HbA2
- C. HbF

74) increased mother serum alpha feto protein (MSAFP) suggests a possibility of

- A. Neural tube defect

68	71	74	
D	B	A	

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Genetics Final Locker21

1- image of infant with down syndrome* What continental disease does he suffer from:

- A- trisomy 18
- B- trisomy 21
- C- trisomy 13

2- 8 year old boy named Ali presented to the clinic with his parents with intellectual disabilities, long face, broad forehead, larger ears protruded mandible. Patient has a similarly affected maternal uncle, what is an indication of this condition?

- A- a karyotype showing 47 XXY result
- B- G6PD enzyme deficiency
- C- abnormal increase in CGG repeats in FMR1 gene

3- Familal microcephaly was diagnosed, it's an Autosomal recessive, a family presented with a 6 years old boy, what should you counsel?

- A- affected children will be 1 to 2
- B- the unaffected sister of the affected boy will be a carrier
- C- the parents are carries is not true

4- which of the following is an indication of prenatal diagnosis?

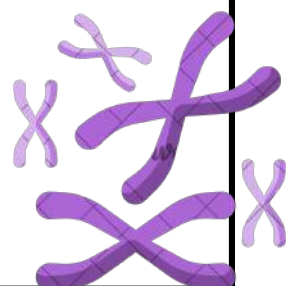
- A- very young maternal age
- B- advanced maternal age
- C- family history of obesity
- D- smoking

6- Which of the following disorders needs a methylation test for conformation?

- A- Achondroplasia
- B- Pader-Will Syndrome
- C- Down Syndrome
- D- Duchenne muscular dystrophy

1	2	3	4	6
B	C	B	B	B

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med 24 final

Q1

What's the best course of action for a 2-week-old neonate with galactosemia?

- A. Lactose-free galactose-free milk
- B. Fructose-free milk
- C. Liver transplant
- D. Stem cell therapy

Q2

Which of the following represents maternal inheritance?

- A. c1878G>T
- B. m.87383>C
- C. p.Arg112Lys
- D. r99C>A

Q3

Which single anomaly birth defect explains a baby being born with congenital heart deformities?

- A. Malformation
- B. Disruption
- C. Deformation
- D. Dysplasia

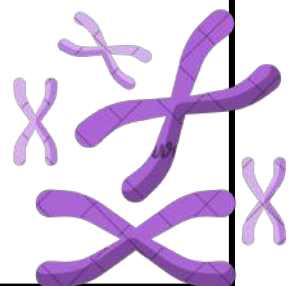
Q4

What is the likelihood of the proband being a carrier?

- A. 25%
- B. 50%
- C. 100%
- D. 1-3%

1	2	3	4
A	B	A	B

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Q5

Newborn presents with vomiting, lethargy, and foul-smelling urine. Laboratory tests reveal elevated phenylalanine. What is the carrier frequency if incidence = 1/10,000?

- A. 1:100
- B. 1:25
- C. 1:50
- D. 1:10

Q6

Neonate presents with vomiting, lethargy, and increased plasma levels of valine, leucine, and isoleucine.

- A. —
- B. Maple Syrup Urine Disease
- C. —
- D. —

Q7

Which chromosomal testing method is most appropriate for detecting a specific genetic disorder caused by a mutation on chromosome 4p?

- A. Sanger sequencing
- B. FISH
- C. Karyotyping
- D. Whole genome sequencing

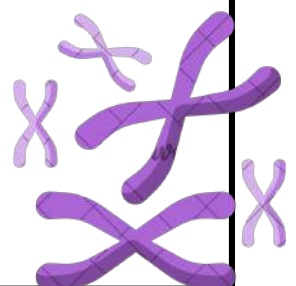
Q8

Which technique is most appropriate to detect a microdeletion in chromosome 4p16.3?

- A. Karyotyping
- B. PCR
- C. FISH
- D. Southern blot

5	6	7	8
C	B	B	C

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Q9

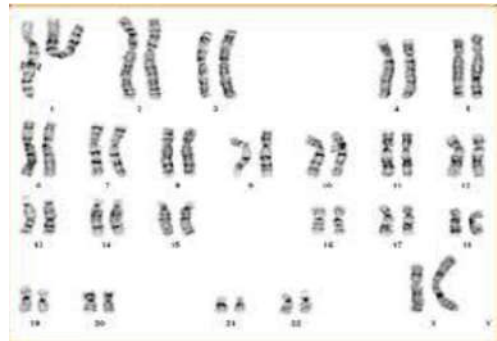
Which displays a frameshift mutation?

- A. c.1258delC (p.Leu1258Phefs)
- B. c.891+1G>A
- C. c.742T>G
- D. c.1003C>A

Q10 

What does this karyotype display?

- A. Normal female
- B. Turner isochromosome
- C. Turner ring chromosome
- D. Turner X deletion



Q11

When identifying a gene associated with breast cancer, which test would be most suitable?

- A. FISH
- B. Gene panel
- C. Karyotyping
- D. Sanger sequencing

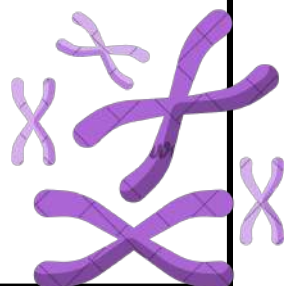
Q12

A 2-year-old child with characteristic facial features and transfusion dependence since infancy. What is the definitive cure?

- A. Chelating agent only
- B. Bone marrow transplant
- C. Hydroxyurea
- D. Constant transfusion

9	10	11	12
A	B	B	B

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Q13

A newborn has Hb Bart's on screening. What hemoglobin is expected?

- A. γ_4
- B. $\alpha_2\beta_2$
- C. $\alpha_2\gamma_2$
- D. $\alpha_2\delta_2$

Q14

Shortened anteroposterior dimension of the head?

- A. Brachycephaly
- B. Dolichocephaly
- C. Plagiocephaly
- D. Trigonocephaly

Q15

Neonate with cleft lip, cleft palate, holoprosencephaly, and congenital heart defects. What is likely also present?

- A. Syndactyly
- B. Polydactyly
- C. Macroductyly
- D. Brachydactyly

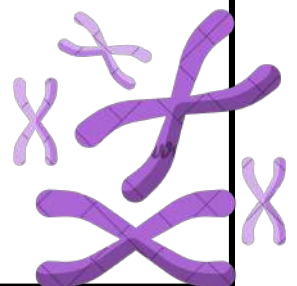
Q16

Patient with short stature, webbed neck, ovarian failure. Responsible gene?

- A. SHOX
- B. FMN1
- C. CFTR
- D. SRY

13	14	15	16
A	A	B	A

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Q17

Cystic fibrosis: what is the correct infertility finding?

- A. Male infertility due to congenital bilateral absence of vas deferens
- B. Male infertility due to testicular atrophy
- C. Female infertility due to —
- D. —

Q18

Mechanism of action of Spinraza (nusinersen)?

- A. Modifying an existing gene's activity
- B. Gene knockdown
- C. Gene editing
- D. Introducing a new gene

Q19

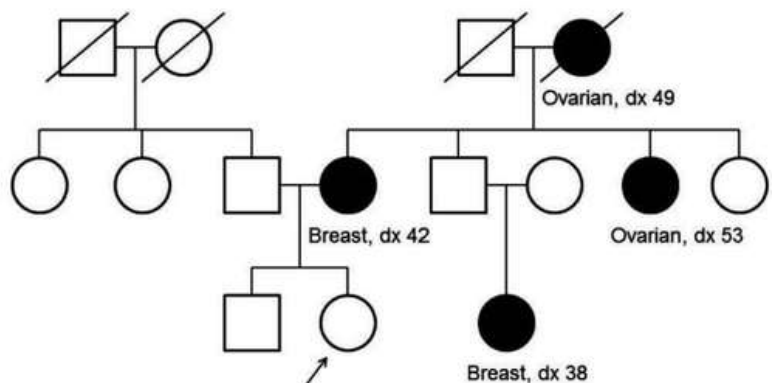
Two affected children with an autosomal dominant disease; both parents unaffected. Most likely explanation?

- A. Variable expressivity
- B. Germline mosaicism
- C. Somatic mosaicism
- D. Two independent de novo mutations

Q20 📷

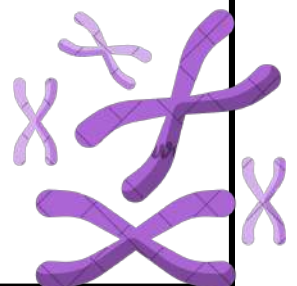
What is the associated gene?

- A. EPCAM
- B. BRCA2
- C. APC
- D. PTEN



17	18	19	20
A	A	B	B

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Q21

Recurrence risk for siblings when a child is born with bilateral cleft lip and palate?

- A. 4.2%
- B. 2.6%
- C. 5.6%
- D. 0.5%

Q22

Gene therapy approach used in Libmeldy for MLD?

- A. Ex-vivo AAV
- B. In-vivo AAV
- C. Ex-vivo Lentivirus
- D. In-vivo Lentivirus

Q23

Gene responsible for the two-hit hypothesis?

- A. Tumor suppressor gene
- B. Growth factor gene
- C. Growth factor receptor gene
- D. Proto-oncogene

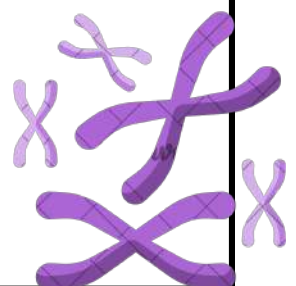
Q24

Which condition is treated with prophylactic total colectomy?

- A. HNPCC
- B. FAP
- C. Lynch syndrome
- D. Bloom syndrome

21	22	23	24
C	C	A	B

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Q25

Which is a qualitative β -globin disorder?

- A. Cooley anemia
- B. Sickle cell anemia
- C. Bart's hydrops fetalis
- D. HbH disease

Q26

Which condition typically requires total colectomy?

- A. Lynch syndrome
- B. FAP
- C. HNPCC
- D. Bloom syndrome

Q27

13-week pregnancy, low PAPP-A and low β -hCG. Best confirmatory test?

- A. Triple test
- B. Quadruple test
- C. CVS
- D. No further testing

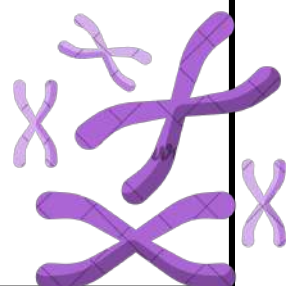
Q28

13-week pregnancy, low PAPP-A and low β -hCG. Most likely condition?

- A. Aneuploidy
- B. Single-gene disorder
- C. Chromosomal abnormality

25	26	27	28
B	B	C	A

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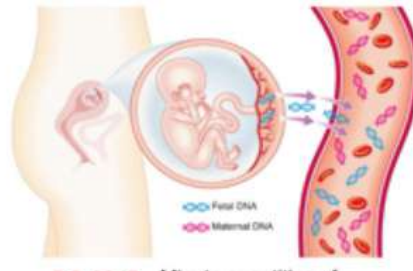


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Q29

What is the best description of the procedure shown?

- A. Chorionic villus sampling**
B. Invasive test
C. Non-invasive test
D. Amniocentesis
E. Cordocentesis



Q30

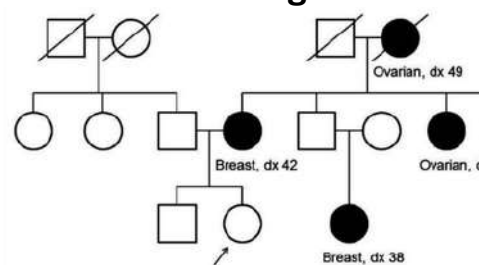
Patient suspected to have sickle cell anemia. Best cytomolecular technique to directly detect HBB mutation?

- A. Spectrophotometry**
B. DNA sequencing
C. CBC
D. Western blot

Q31

What is the recurrence risk of the index case being a carrier?

- A. 100%**
B. 50%
C. 1-2% or 2-3%
D. 25%



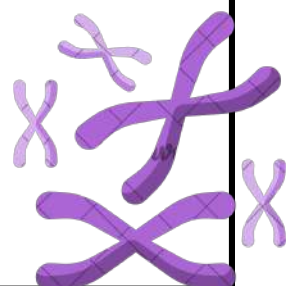
Q32

13-week pregnancy, low PAPP-A and low β -hCG, suspected trisomy. Best confirmatory test?

- A. Quadruple test**
B. Triple test
C. CVS
D. Screening is sufficient

29	30	31	32
C	B	B	C

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Q33

2-year-old child with intellectual disability and hair hypopigmentation.

- A. PKU
- B. Galactosemia
- C. Homocystinuria
- D. Tay-Sachs

Q34

Which mutation type is controversial in gene therapy?

- A. Somatic mutation
- B. Germline mutation
- C. Ex-vivo therapy
- D. Anti-sense mutation

Q35

Mutation causing malignant hyperthermia?

- A. RYR1
- B. pCHE
- C. TPMT
- D. DPYD

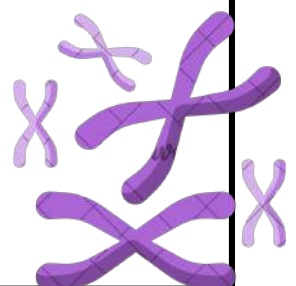
Q36

Deletion of short arm of chromosome 10 in a male?

- A. 46,XX,del(10)(p11.2p23)
- B. 46,XY,del(10)(p11.2p23)
- C. 45,XY,del(10)(q11.2q23)
- D. 46,XY,del(10)(p11q23)

33	34	35	36
A	B	A	B

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Q37

Red-green color blindness inheritance?

- A. Autosomal dominant
- B. Autosomal recessive
- C. X-linked recessive
- D. X-linked dominant

Q38

Marfan syndrome is an example of:

- A. Pleiotropy
- B. Codominance
- C. Autosomal dominant inheritance
- D. Polygenic inheritance

Q39

Condition commonly associated with cleft lip and palate in trisomy 13?

- A. Clinodactyly
- B. Syndactyly
- C. Polydactyly
- D. Micrognathia

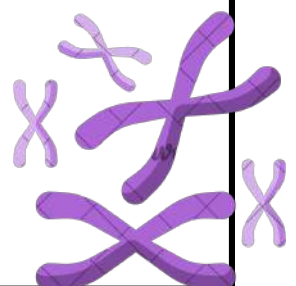
Q40

Lynch syndrome gene?

- A. APC
- B. BRCA1
- C. MSH1/MSH2

37	38	39	40
C	A	C	C

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med 24 final

Q41

Which of the following represents a pericentric inversion in a female?

- A. 46,XX,inv(6)(p15p18)
- B. 46,XX,inv(6)(q15q18)
- C. 46,XX,inv(6)(p15q18)
- D. 46,XX,i(6)(p21p23)

Q42

Which is a clinical feature of homocystinuria?

- A. Sweet urine odor
- B. Dark urine
- C. Lens dislocation
- D. —

Q43

Which of the following is the mechanism of DNA repair?

- A. Cleavage → Removal → Insertion → Sealing → Repair
- B. Removal → Cleavage → Insertion → Sealing → Repair

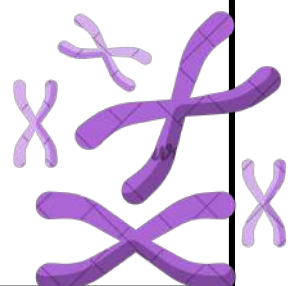
Q44

Molecular mechanism underlying familial hypercholesterolemia?

- A. Haploinsufficiency
- B. Dominant negative mutation
- C. Gain of function

41	42	43	44
C	C	A	A

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med 24 final

45

Characteristic clinical feature of PKU?

- A. Hair hypopigmentation
- B. Sweet-smelling urine

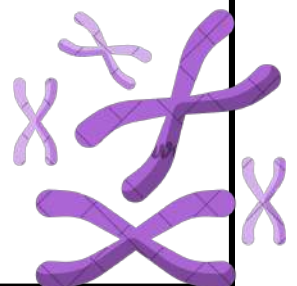
Q46

One parent affected with sickle cell disease (SS) and the other has sickle cell trait (AS). Recurrence risk?

- A. 50% affected child
- B. 25% unaffected child
- C. No counseling needed

45	46		
A	A		

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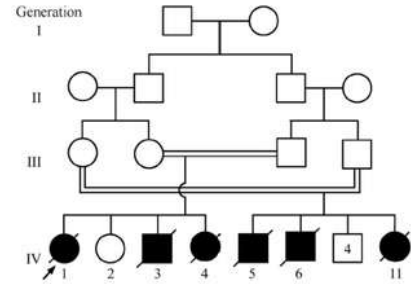


med18 final

Q1

This pedigree shows what mode of inheritance?

- A. Autosomal dominant
- B. Autosomal recessive
- C. X-linked recessive
- D. Mitochondrial



Q2

A 6-year-old boy presented with hypogonadism, obesity, and almond-shaped eyes. What is the diagnosis?

- A. Down syndrome
- B. Fragile X syndrome
- C. Angelman syndrome
- D. Prader-Willi syndrome

Q3

Which enzyme converts RNA to DNA?

- A. Reverse decarboxylase
- B. DNA polymerase
- C. Reverse transcriptase

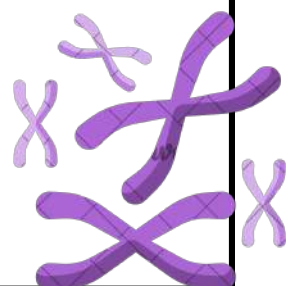
Q4

A male with tuberculosis is using isoniazid and develops a severe adverse effect. What should be done?

- A. Increase the dose
- B. Continue treatment without stopping
- C. Change the drug
- D. Add another drug

1	2	3	4
B	D	C	C

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med18 final

Q5

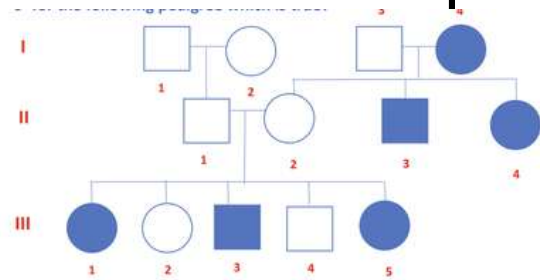
A couple has a child with bilateral cleft lip and palate. What is the recurrence risk for another affected child?

- A. 5.6%
- B. 2%
- C. 50%
- D. 4%

Q6

For the following pedigree, which statement is true?

- A. Individual III-5 is a deceased affected female
- B. Individual II-2 might exhibit reduced penetrance
- C. The mode of inheritance is X-linked recessive
- D. Individual I-2 is an affected female



Q7

Medication recommended during the first trimester of pregnancy?

- A. Streptomycin
- B. Folic acid
- C. Isoniazid

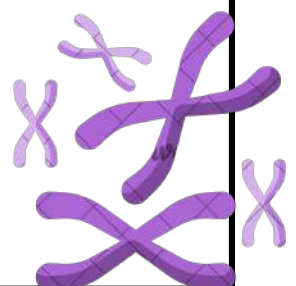
Q8

Two parents have two affected children with Hemophilia A. They want a healthy child. What do you recommend?

- A. Preimplantation genetic diagnosis
- B. Newborn screening
- C. Early diagnosis

5	6	7	8
A	B	B	A

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med18 final

Q9

Mutations in BRCA2 gene:

- A. Account for 3–5% of familial breast cancer
- B. Account for 100% of familial breast cancer
- C. Males have no breast cancer risk
- D. Males have increased prostate cancer risk

Q10

What is a potential reason for pediatric genetic counseling?

- A. Adult with family history of a genetic condition
- B. Adult with family history of adult-onset disorders
- C. Adult with family history of cancer
- D. Child with family history of a genetic condition

Q11

An increase in inhibin-A in maternal serum is associated with:

- A. Cystic fibrosis
- B. Down syndrome
- C. Turner syndrome
- D. Duchenne muscular dystrophy

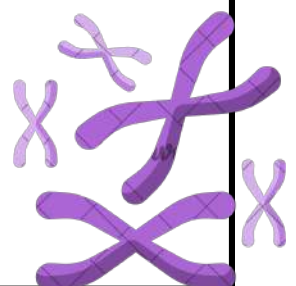
Q12

A mother has one child with galactosemia and is pregnant in her second trimester. What test is advised?

- A. Newborn screening
- B. Parental carrier testing
- C. Ultrasonography

9	10	11	12
D	D	B	B

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med18 final

Q13

According to genetic counseling, all are true EXCEPT:

- A. Genetic counseling is directive
- B. Any physician can be a counselor
- C. Includes discussion of disease cause and recurrence risk
- D. Should be done after genetic testing

Q14

Which is considered a major congenital anomaly in Prader-Willi syndrome?

- A. Hypogonadism
- B. Polydactyly
- C. Almond-shaped eyes
- D. Upward slanting eyes

Q15

PCR amplifies DNA using:

- A. DNA polymerase
- B. Restriction enzyme
- C. Plasmid
- D. Phages

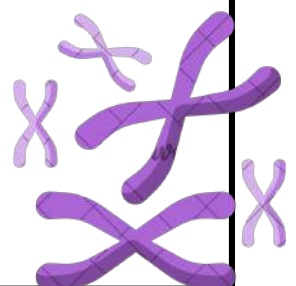
Q16

Which DNA repair mechanism is a direct reversal mechanism?

- A. Post-replication repair
- B. Excision repair
- C. Mismatch repair
- D. Photoreactivation

13	14	15	16
A	A	A	D

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Q17

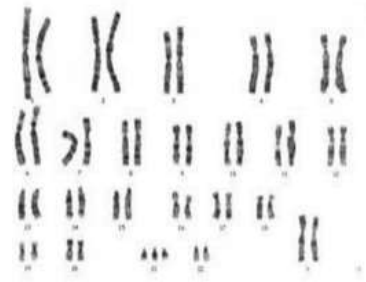
Mutation in all four alpha-globin alleles is associated with:

- A. HbH disease
- B. Hydrops fetalis
- C. Cooley anemia

Q18 

A couple has a baby with a genetic disorder confirmed by this karyotype. What is the recurrence risk?

- A. Related to maternal age
- B. 1–3% if transmitted from father
- C. 10–15% if transmitted from mother
- D. 100%



Q19

A 30-year-old male has chronic pulmonary disease and infertility. Diagnosis?

- A. Klinefelter syndrome
- B. Cystic fibrosis
- C. Marfan syndrome
- D. Prader-Willi syndrome

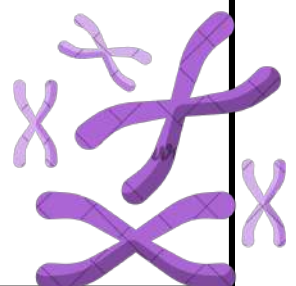
Q20

Which prenatal test does NOT carry a risk of fetal loss?

- A. Maternal serum AFP
- B. Chorionic villus sampling
- C. Fetal tissue sampling
- D. Amniocentesis

17	18	19	20
B	A	B	A

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med18 final

Q21

NAT2 gene is related to which drug?

- A. Succinylcholine
- B. G6PD deficiency
- C. Isoniazid

Q22

Why are stem cells desirable therapeutically?

- A. Become nerve cells
- B. Found in reproductive tissue
- C. Differentiate into many tissue types
- D. Form whole organs

Q23

Detection of proteins by Western blot relies on:

- A. Protein-antibody specificity
- B. Labeling antibody with dye
- C. Protein denaturation
- D. Labeling protein with dye

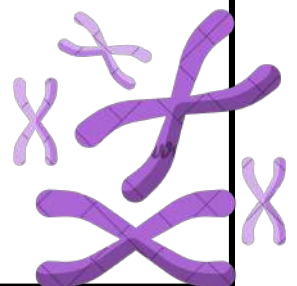
Q24

A child with SCID treated using retroviral gene therapy develops leukemia-like disease. Cause?

- A. Insertion near proto-oncogene promoter
- B. SCID itself
- C. Triggered HTLV
- D. Viral contamination

21	22	23	24
C	C	B	A

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Q25

A 2-month-old baby has normal blood glucose. Which disorder is most likely?

- A. Mucopolysaccharidosis
- B. PKU
- C. Urea cycle defect

Q26

Boy with almond-shaped eyes and obesity. Diagnosis?

- A. Prader-Willi syndrome
- B. Angelman syndrome
- C. PKU

Q27

Arachnodactyly is characteristic of:

- A. Turner syndrome
- B. Marfan syndrome
- C. Patau syndrome
- D. Cri-du-chat syndrome

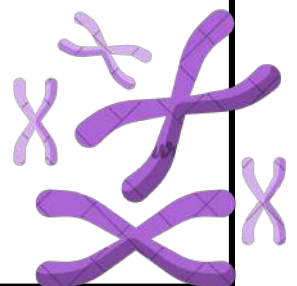
Q28

Potter sequence occurs due to:

- A. External disruption
- B. Renal agenesis
- C. Intrauterine infection
- D. Placental dysfunction

25	26	27	28
B	A	B	B

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Q29

Minor congenital anomaly in Down syndrome:

- A. VSD
- B. Congenital hypothyroidism
- C. Intellectual disability
- D. Clinodactyly

Q30

Which condition can be diagnosed using plasma amino acids?

- A. Zellweger syndrome
- B. Maple syrup urine disease
- C. Galactosemia
- D. Fatty acid oxidation disorder

Q31

Which statement about inherited cancer is TRUE?

- A. Cancer is usually inherited
- B. Parent may carry mutation and remain unaffected while child develops cancer
- C. Cancer never skips generations
- D. Familial cancers are caused by normal genes

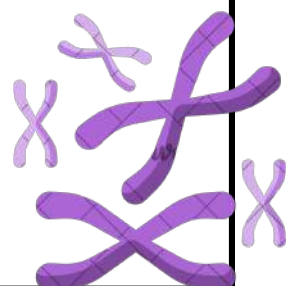
Q32

Feature suggestive of inherited cancer syndrome:

- A. Several close relatives with same cancer
- B. One relative with rare cancer
- C. Late age of onset
- D. Unilateral tumor

29	30	31	32
D	B	B	A

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Q33

Newborn with vertebral anomalies, anal atresia, tracheoesophageal fistula, and radial ray defects:

- A. Ectodermal dysplasia
- B. VATER association
- C. Potter syndrome
- D. Van der Woude syndrome

Q34

DMD gene is >2000 kb but mRNA is only 14 kb. Why?

- A. Exons removed
- B. More amino acids coded in DNA
- C. DNA is double stranded
- D. Introns removed

Q35

A 16-year-old female with short stature and primary amenorrhea. Chromosomal analysis shows:

- A. 45,XX,t(14;21)
- B. 45,XX,rob(21;22)
- C. 46,X,isoXq

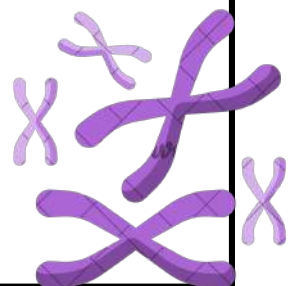
Q36

Major abnormality in Patau syndrome:

- A. Cleft lip/palate and microcephaly
- B. Polydactyly
- C. Macrocephaly

33	34	35	36
B	D	C	A

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med18 final

Q37

A vector was inserted and later caused adverse effects. Most likely vector?

- A. Retrovirus
- B. Adenovirus
- C. Herpes simplex virus

Q38

A patient is a sickle cell trait carrier. Which statement is correct?

- A. Homozygous autosomal dominant disorder
- B. Autosomal recessive and protection against malaria
- C. X-linked recessive disorder

Q39

A 2-year-old girl presents with hepatomegaly and fasting hypoglycemia. Diagnosis?

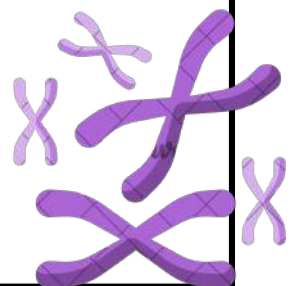
Q40

A newborn in NICU has convulsions, hypoglycemia, and absent ketone bodies.

- A. MSUD
- B. Glycogen storage disease
- C. Hereditary galactosemia

37	38	39	40
B	B		B

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med18 final

Q41

Regarding genetic counseling, which is true?

- A. Only after testing
- B. Directive
- C. Any physician can do it
- D. Includes discussion of disease cause and transmission

Q42

Consanguineous couple with family history of PKU. Appropriate counseling?

- A. AD with reduced penetrance
- B. AR; 50% chance of being carriers
- C. Multifactorial; 4% risk
- D. X-linked recessive

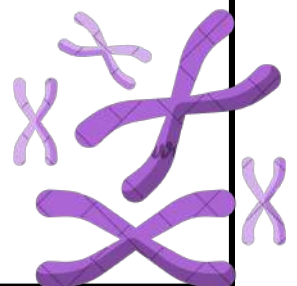
Q43

Triple screen includes all EXCEPT:

- A. AFP
- B. hCG
- C. Unconjugated estriol
- D. Indirect bilirubin

41	42	43	
D	B	D	

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IEM MED 19

Q1

A child presented with non-ketotic hypoglycemia, this may be due to:

- a) Hereditary fructose intolerance
- b) Hyperinsulinism
- c) Von Gierke Disease
- d) Galactosemia

Q2

A newborn infant presented with hypotonia and weakness and has mildly dysmorphic facial features consisting of a prominent forehead and a large anterior fontanel, and an enlarged liver. Investigations revealed renal cysts and increased plasma very long chain fatty acids. The pathogenesis of such disorder is due to:

- a) Severely reduced numbers of peroxisomes in all cells.
- b) Abnormal accumulation of one amino acid proximal to a metabolic block.
- c) Deficiency of a lysosomal enzyme causing accumulation of macromolecules.
- d) Inborn errors of glycogen synthesis and degradation enzymes.

Q3

A 6-year-old female presented with intellectual disability, seizures, fair hair and skin, and a musty odor of sweat and urine. In this disorder:

- a) Treatment involves a low-methionine diet with cystine supplementation.
- b) There is deficiency of the enzyme required for conversion of phenylalanine to tyrosine.
- c) Dietary fructose restriction gives a good prognosis.
- d) Diagnosis is confirmed by branched-chain amino acids in blood.

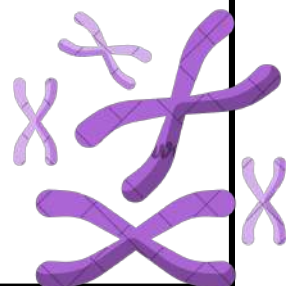
Q4

A neonate admitted to the NICU with ammonia level above 200 $\mu\text{mol/L}$. Which disorder is suspected?

- a) Glycogen storage disease
- b) Urea cycle disorder
- c) Mitochondrial disorder
- d) Mucopolysaccharidosis

1	2	3	4
B	A	B	B

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IEM MED 19

Q5

A 4-month-old male presented with jaundice since birth, cataract, and hepatomegaly. Investigations revealed reducing substances in urine.

Management should be:

- a) Prescribing lactose-free milk formula
- b) Giving tyrosine supplements
- c) Restricting leucine and valine
- d) Restricting fructose

Q6

A 3-month-old girl with abdominal distension and jaundice is on breastfeeding. Her labs revealed ketotic hypoglycemia. Based on the management plan:

- a) Fructose-free milk
- b) Low-fat milk
- c) Frequent feeding with uncooked starch
- d) Low-galactose diet

Q7

The laboratory reports significant elevation of plasma arginine. What additional abnormality is expected?

- a) Hyperammonemia
- b) Lactic acidosis
- c) Hypoglycemia
- d) Elevated muscle enzymes

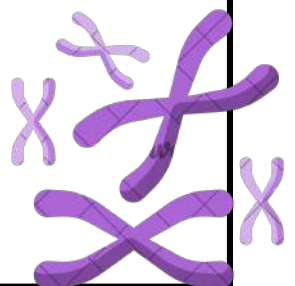
Q8

A 1-year-old boy presents with intellectual disability, fair skin, and unusual urine odor. What is the disease mechanism?

- a) Energy failure
- b) Accumulation of complex molecules
- c) Excess of essential metabolites
- d) Accumulation of toxic metabolites

5	6	7	8
A	D	A	D

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IEM MED 19

Q9

Choose the correct matching (Disease – Diagnostic biochemical test):

- a) Mucopolysaccharidosis – Ammonia
- b) Homocystinuria – Plasma Methionine
- c) Zellweger syndrome – Plasma Glucose
- d) Galactosemia – Lactic acid

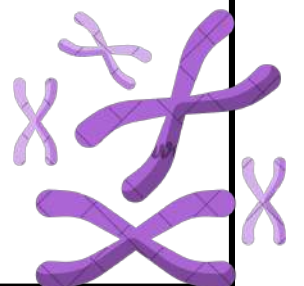
Q10

Plasma amino acid analysis is a useful diagnostic tool for which disorder?

- a) Fatty acid oxidation defect
- b) Galactosemia
- c) Maple syrup urine disease
- d) Organic acidemia

9	10		
B	C		

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Med 18, 5th ps

Q1

The presence of an extra digit is called:

- A. Arachnodactyly
- B. Polydactyly
- C. Syndactyly
- D. Brachydactyly

Q2

Results from external factors disrupting the normal developmental process:

- A. Deformation
- B. Association
- C. Disruption
- D. Dysplasia

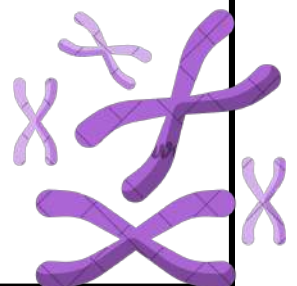
Q3

A family comes with a newborn baby with a protruding mass in the lower back, weakness of both limbs, and club feet. They previously had a similarly affected baby. What is the recurrence risk for the next baby?

- A. 4%
- B. 1%
- C. 0.5%
- D. 10%

1	2	3	
B	C	D	

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PS5

1. Abnormal decrease in the intraorbital distance is termed hypotelorism.

- A. True
- B. False

2. Werdnig-Hoffmann disease is due to exon 7 deletion within SMN1 gene.

- A. True
- B. False

3. Microcephaly is considered if the head circumference is above +3 standard deviations.

- A. True
- B. False

4. Ectodermal dysplasia resulting from an abnormal mechanical force that distorts a normal structure.

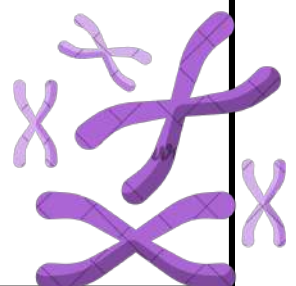
- A. True
- B. False

5. A patient has aortic aneurysm and arachnodactyly. Which syndrome is most likely?

- A. Down syndrome
- B. Marfan syndrome
- C. Prader-Willi syndrome
- D. Turner syndrome

1	2	3	4	5
A	A	B	F	B

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PS5

1) A boy 2 months old with dysmorphic features, severe Hypotonia, as well as liver and renal cysts?

- a- DNA analysis of SMNI gene.
- b- DNA analysis of Dystrophin gene.
- c- Chromosomal studies.
- d- Evaluation of serum very long chain fatty acids.

2) A 7-year-old boy has chronic cough, diarrhea, and malnutrition. Which investigation is most appropriate?

- A. DNA analysis of SMN1 gene
- B. DNA analysis of CFTR gene
- C. Chromosomal studies
- D. DNA analysis of FGFR3 gene

3) An 8-year-old girl presents with short stature, webbed neck, and increased carrying angle. Which investigation is most appropriate?

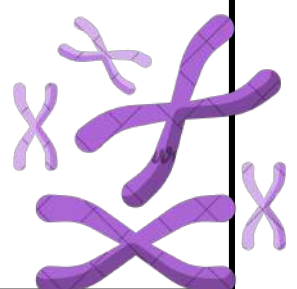
- A. DNA analysis of Dystrophin gene
- B. DNA analysis of CFTR gene
- C. Chromosomal studies
- D. DNA analysis of FGFR3 gene

4) A patient has duodenal atresia and a wide gap between the 1st and 2nd toes. Which syndrome is most likely?

- A. Marfan syndrome
- B. Prader-Willi syndrome
- C. Down syndrome
- D. Neurofibromatosis

1	2	3	4
D	B	C	C

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Problem solving

1) Which of the following is CORRECT about germ line gene therapy?

- A. Genetic changes will not be inherited by future generations.
- B. Currently the only gene therapy ethically considered.
- C. Gametes are the only targets for this type of gene therapy.
- D. All cells except those in the germ-line are potential targets.

2) All the following is correct for somatic cell gene therapy EXCEPT:

- A. Muscle cells can be used as targets.
- B. Liver cells can be used as a target.
- C. Nerve cells can be used as a target.
- D. Genetic changes would be transmitted to future generations.

3-Ex vivo gene therapy:

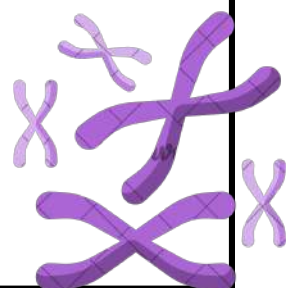
- A. The removal of cells from the affected individual, modifying them genetically, and then retune them to the body.
- B. The infusion of adenoviral vectors into the trachea and bronchi of patients with cystic fibrosis.
- C. The insertion of therapeutic gene into a plasmid vector.
- D. Selection of transformed cells before returning them into the patient.

4- A patient has hypogonadism, cryptorchidism, and almond-shaped eyes. Which syndrome is most likely?

- A. Down syndrome
- B. Marfan syndrome
- C. Prader-Willi syndrome
- D. Neurofibromatosis

1	2	3	4
C	D	A	C

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Problem solving

1) Oncogenic viral vectors are effective means of gene therapy in mammalian cells. However, their use is hindered by:

- A. Can be used only for ex vivo gene therapy,
- B. Infecting a wide spectrum of cells,
- C. Insertional mutagenesis
- D. The foreign gene is transiently expressed.

2) Adenoviral vectors are effective means for gene therapy, However they have the disadvantage of the:

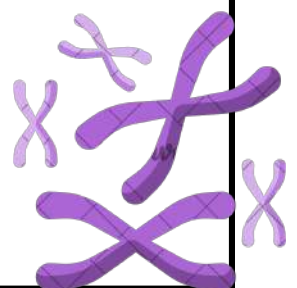
- A. Requirement of co-infection with a helper virus.
- B. Requirement of cell division.
- C. unstable expression of the introduced gene.
- D. integration in host genome.

Sustained correction of X- severe combined immunodeficiency (SCID) due to ADA deficiency has been observed in clinical trials. Cells in patients with ADA deficiency are treated by lymphocyte gene therapy, persists and maintain transgene expression for many years. In 2002 two of ten children with SCID treated by gene therapy at the trial's Paris center developed a leukemia-like condition due to inappropriate insertion of the retroviral vector near a proto-oncogene promoter led to uncontrolled clonal proliferation of mature T-cells.

- 1) What type of gene therapy used in this trail? **Somatic Gene therapy**
- 2) What is the route used of gene therapy used in this trail? **Ex-vivo**
- 3) What is the type of vector used in this trail? **Viral vector**
- 4) What is the cause of developing a leukemia-like condition? **Insertional mutagenesis**
- 5) Mention another disadvantage of this vector? **Limited copying capacity**

1	2	3	4
C	C		

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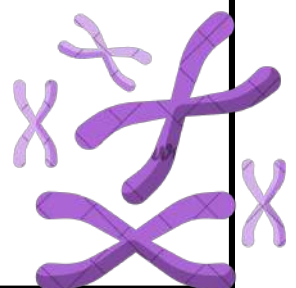
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1	2	3	4
C	C		

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Genetics Final

1. Which chromosome is most commonly involved in Robertsonian translocation, causing Down syndrome?

- A- Chromosome 15
- B- Chromosome 3
- C- Chromosome 1
- D- Chromosome 14

2. Which of the following karyotypes is an example of anaphase lag?

- A- 47, XY, +21
- B- 45, X
- C- 69, XXY
- D- 46, XX, del(5p)

3. A chromosome is divided transversely through the centromere. Which chromosomal abnormality results?

- A- Isochromosome
- B- Ring chromosome
- C- Paracentric inversion
- D- Pericentric inversion

4. A woman with Neurofibromatosis Type 1 marries an unaffected man. Which statement regarding their children is correct?

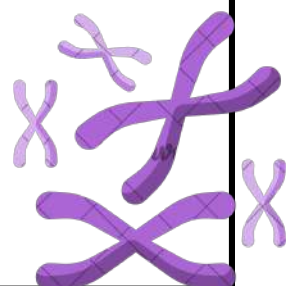
- A- The probability that the third child will be affected, even if the first two children are affected, is 50%.
- B- The probability that the third child will be affected is 25%.
- C- All children will be affected.
- D- No children will be affected.

5. In the pedigree shown, individual II-2 is:

- A- Affected female
- B- Unaffected female
- C- Carrier female
- D- Deceased female

1	2	3	4	5
D	B	A	A	A

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Genetics Final

6. Which disorder may result from uniparental disomy?

- A- Turner syndrome
- B- Albinism
- C- Angelman syndrome
- D- Myotonic dystrophy

7. Deficiency of which gene increases the risk of severe toxicity from 5-fluorouracil (5-FU)?

- A- BCHE (pCHE)
- B- CYP2D6
- C- DPYD
- D- TPMT

8. Favism is associated with mutations in which gene?

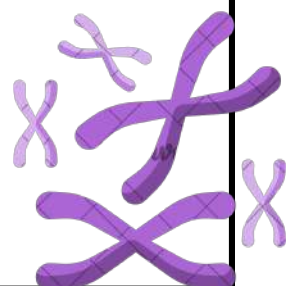
- A- G6PD
- B- CYP2C9
- C- NAT2
- D- TPMT

9. Quantitative hemoglobin analysis showing a marked increase in HbF and a slight increase in HbA₂ is most characteristic of:

- A- Sickle cell anemia
- B- Hereditary persistence of fetal hemoglobin (HPFH)
- C- β -thalassemia
- D- α -thalassemia

6	7	8	9
C	C	A	C

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Genetics Final

10. A patient presents with intellectual disability, developmental delay, and dysmorphic features without a specific clinical diagnosis. What is the most appropriate diagnostic approach?

- A- Chromosome analysis → Chromosomal microarray → Whole-exome sequencing (WES)
- B- Chromosome analysis → Sanger sequencing → WES
- C- Chromosome analysis → Chromosomal microarray → MLPA
- D- Chromosome analysis → FISH → WES

11. A 37-year-old woman with a BRCA1 mutation develops breast cancer, while her mother carries the same mutation but remains unaffected. What is the best explanation?

- A- Incomplete penetrance
- B- Autosomal dominant inheritance
- C- BRCA1 is a tumor suppressor gene
- D- Variable expressivity

12. What genetic abnormality is demonstrated by this result?

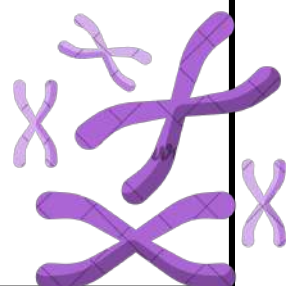
- A- Heterozygous microdeletion
- B- Homozygous duplication
- C- Heterozygous duplication
- D- Homozygous microdeletion

13. A couple has a child with Duchenne muscular dystrophy (DMD). Which test is most appropriate to determine whether another family member carries the mutation?

- A- Sanger sequencing
- B- MLPA
- C- Karyotyping
- D- FISH

10	11	12	13
A	A	A	B

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Genetics Final

14. An ammonia level of 400 $\mu\text{mol/L}$ in a neonate is most characteristic of:

- A- Phenylketonuria
- B- Homocystinuria
- C- Hyperargininemia
- D- Galactosemia

15. Which molecular technique is most commonly used to detect exon deletions and duplications in Duchenne muscular dystrophy?

- A- MLPA
- B- Karyotyping
- C- FISH
- D- PCR-RFLP

16. Favism is caused by:

- A- Mutation in the G6PD gene
- B- Mutation in CYP2C9
- C- Mutation in NAT2
- D- Mutation in TPMT

17. Hereditary Nonpolyposis Colorectal Cancer (HNPCC/Lynch syndrome) is associated with mutations in:

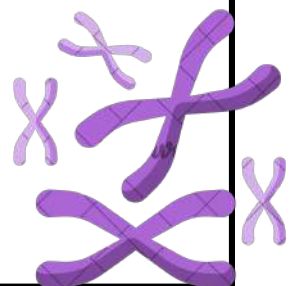
- A- APC and KRAS
- B- BRCA1 and BRCA2
- C- MSH2 and MLH1
- D- RB1 and TP53

18. Talk in details

- A- About investigation
- B- Symptoms

14	15	16	17	18
C	A	A	C	A

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Genetics Final

19. A patient is diagnosed with Lynch syndrome due to an MLH1 mutation. Which test is most appropriate for predictive testing of an asymptomatic sibling?

- A- Gene panel
- B- FISH analysis
- C- Sanger sequencing
- D- MLPA

20. If HNPCC (Lynch syndrome) is confirmed by genetic testing, what is the most appropriate next step in management?

- A- Enroll the patient in an annual screening and surveillance program
- B- Refer the patient to a genetic counselor
- C- Refer the patient for subtotal colectomy
- D- Refer the patient for a colonoscopy

21. Case ...

- A- preplantation
- B- newborn ...

22. Which two major congenital anomalies are commonly associated with Down syndrome?

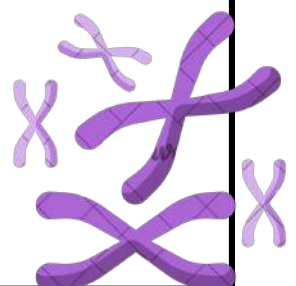
- A- Microcephaly and polydactyly
- B- Duodenal atresia and atrial septal defect
- C- Cleft palate and spina bifida
- D- Renal agenesis and syndactyly

23. Premarital genetic screening can be used to detect which of the following disorders?

- A- Down syndrome
- B- Sickle cell disease
- C- Marfan syndrome
- D- Balanced chromosomal translocation

19	20	21	22	23
C	A	A	B	B

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Genetics Final

24. Which type of viral vector was used in the COVID-19 vaccine?

- A- Adenovirus vector
- B- Lipid nanoparticle (mRNA vaccine)
- C- Retrovirus vector
- D- Lentivirus vector

25. about Corona virus

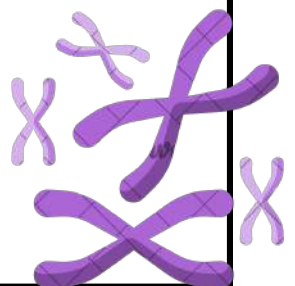
- A- somatic

26. What is the main advantage of this vector?

- A- Can carry a large DNA insert
- B- Long duration of expression
- C- Low transfection efficiency
- D- High immunogenicity

24	25	26
B	A	A

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Genetics Final Med 15

1. Consanguineous parents have a high risk of having a child with which of the following disorders?

- A- Cystic fibrosis
- B- G6PD deficiency
- C- Hemophilia

2. PCR amplification of DNA requires:

- A- Restriction endonuclease
- B- Heat-stable DNA polymerase
- C- Single-stranded DNA probes
- D- Labeled antibodies

3. Which of the following represents the karyotype of a female with translocation Down syndrome?

- A- 46,XX,rob(14;21),+21
- B- 47,XXY
- C- 46,XX,rob(13;15),+21
- D- 46,XY,der(12;21),+12

4. Genetic counseling consists of all the following steps EXCEPT:

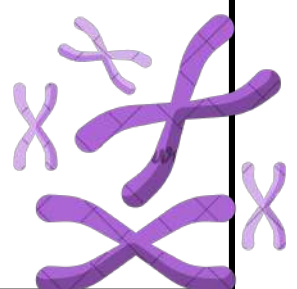
- A- Diagnosis
- B- Risk assessment
- C- Communication
- D- Discussion of options
- E- Short-term contact

5. All of the following are tumor suppressor genes EXCEPT:

- A- H-RAS
- B- APC
- C- p53
- D- BRCA1
- E- BRCA2

1	2	3	4	5
A	B	A	E	A

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Genetics Final Med 15

6. A 6-year-old girl presents with hyperphagia, obesity, short stature, almond-shaped eyes, and small hands and feet. What is the most likely diagnosis?

- A- Angelman syndrome
- B- Prader-Willi syndrome
- C- Turner syndrome
- D- Klinefelter syndrome

7. An important laboratory finding associated with acute encephalopathy in a urea cycle disorder is:

- A- Hypercalcemia
- B- Hyperammonemia
- C- Hyperbilirubinemia
- D- Hypoglycemia

8. The addition of a poly-A tail to mRNA functions to:

- A- Protect mRNA from degradation
- B- Facilitate passage of DNA through the nucleus
- C- Bind to the promoter region

9. Mutation of the Sonic Hedgehog (SHH) gene can lead to:

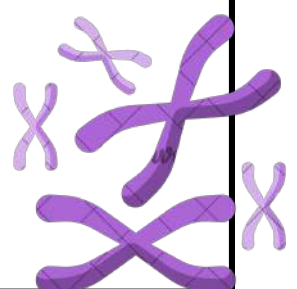
- A- Achondroplasia
- B- Down syndrome
- C- Holoprosencephaly
- D- Turner syndrome

10. Which of the following disorders is an example of a gene-environment interaction?

- A- Sickle cell anemia
- B- Angelman syndrome
- C- Spina bifida
- D- Fragile X syndrome
- E- Neurofibromatosis

6	7	8	9	10
B	B	A	C	C

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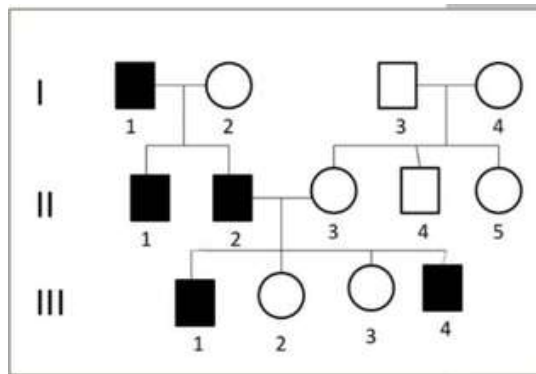
Genetics Final Med 15

11. Which medication is recommended during the first trimester of pregnancy?

- A- Warfarin
- B- Folic acid
- C- Streptomycin
- D- Phenytoin

12. What is the mode of inheritance shown in the pedigree?

- A- Y-linked (Holandric)
- B- Autosomal recessive
- C- X-linked recessive
- D- Autosomal dominant
- E- X-linked dominant



13. Decreased distance between the orbits is called:

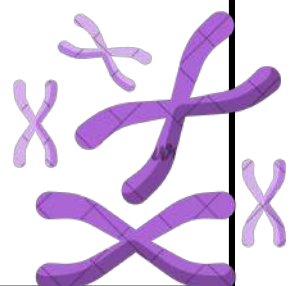
- A- Hypotelorism

14. Premarital screening tests couples for genetic, infectious, and blood-borne diseases. All of the following are included EXCEPT:

- A- Sickle cell anemia
- B- Thalassemia
- C- Antibody screening
- D- PKU
- E- HIV

11	12	13	14
B	A	A	D

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Genetics Final Med 15

15. All cells contain the same genetic information. Why can cells other than stem cells not differentiate into various tissues?

- A- Stem cells have a haploid chromosome number
- B- As cells develop, some genes are permanently turned off
- C- Stem cells are the only cells that can be implanted
- D- As cells develop, their genetic makeup changes
- E- Stem cells are the only cells that do not have X or Y chromosomes and can therefore become either male or female

16. Deficiency of the branched-chain α -ketoacid dehydrogenase complex causes:

- A- Maple syrup urine disease
- B- Classic galactosemia
- C- Phenylketonuria
- D- Cerebrohepatorenal syndrome
- E- Citrullinemia

17. An affected male transmits a disease to all of his daughters but none of his sons. What is the mode of inheritance?

- A- X-linked recessive
- B- X-linked dominant
- C- Mitochondrial inheritance
- D- Autosomal dominant

18. PCR depends on the use of:

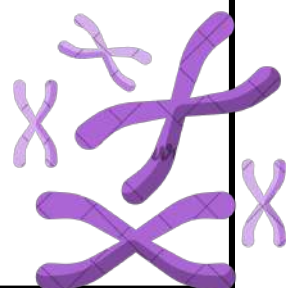
- A- Heat-stable DNA polymerase

19. A male patient presents with almond-shaped eyes, small hands and feet, hyperphagia, obesity, and short stature. What is the most likely diagnosis?

- A- Prader-Willi syndrome
- B- Down syndrome
- C- Turner syndrome
- D- Klinefelter syndrome

15	16	17	18	19
B	A	B	A	A

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Genetics Final Med 15

20. Which of the following is a non-invasive prenatal diagnostic method?

- A- Amniocentesis
- B- Umbilical blood sampling
- C- 3D ultrasound
- D- Embryoscopy
- E- Chorionic villus sampling

21. What is the most common cause of cervical cancer?

- A- Human papillomavirus (HPV)
- B- Contraceptive drugs
- C- Multiple pregnancies

22. Polydactyly is a characteristic feature of:

- A- Patau syndrome
- B- Cri-du-chat syndrome
- C- Achondroplasia
- D- Down syndrome

23. A karyotype showing a ring X chromosome is most consistent with:

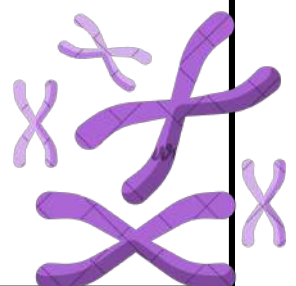
- A- Down syndrome
- B- Klinefelter syndrome
- C- Turner syndrome

24. Electroporation facilitates the introduction of DNA into a target organism by:

- A- Lysis of the cell wall
- B- Lipid-mediated transport
- C- Alteration of cell wall receptors
- D- Formation of temporary pores in the cell membrane

20	21	22	23	24
C	A	A	C	D

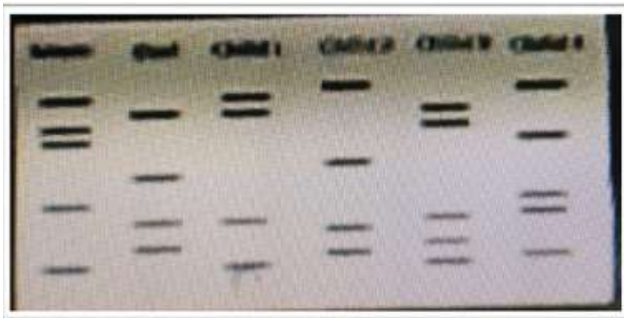
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Genetics Final Med 15

30. Mr. Majed's family consists of a mother, father, and four children. The parents have one son and one daughter together, another daughter is from the mother's previous marriage, and one son is adopted. Based on the DNA profile, which statement is correct?

- A- Children 1 and 3 are siblings
- B- Child 4 is adopted
- C- Child 2 is the mother's daughter from a previous marriage
- D- Children 2 and 3 are siblings
- E- Child 1 is adopted



For question 30

31. Which viral vector is preferred for the treatment of neurological disorders such as Parkinson's disease?

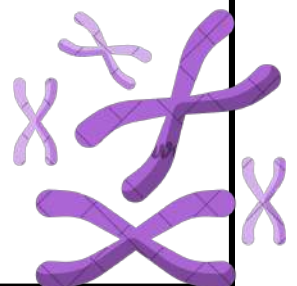
- A- Vaccinia virus
- B- Adenovirus
- C- Adeno-associated virus (AAV)
- D- Herpes simplex virus (HSV)

32. A 12-year-old boy presents with tall stature, arm span greater than height, and anterior lens dislocation. To confirm the diagnosis, which test should be ordered?

- A- FISH for chromosome 15q11-q13 deletion
- B- Measurement of very-long-chain fatty acids
- C- Molecular analysis for CTG trinucleotide repeats
- D- DNA analysis for FGFR3 gene
- E- Molecular analysis of the FBN1 gene

30	31	32
A	D	E

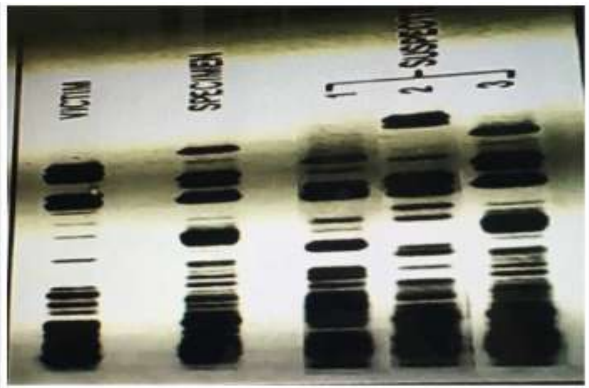
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Genetics Final Med 15

33. A blood sample was collected from a crime scene. DNA samples from the victim and three suspects were analyzed. Based on the DNA profile, which statement is correct?

- A- None of the suspects is the criminal
- B- Suspects 2 and 3 can both be excluded
- C- Suspects 1 and 2 could both be the criminals
- D- Suspect 3 is the criminal



For question 33

34. Gene therapy used to stop or block the expression of a specific gene is called:

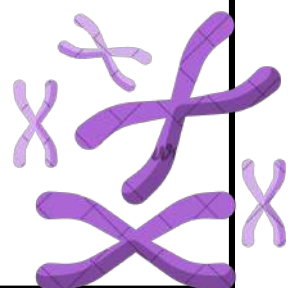
- A- Somatic gene therapy
- B- Antisense therapy
- C- Ex vivo gene therapy

35. A 12-year-old boy from Egypt presents with facial abnormalities, splenomegaly, and hematologic abnormalities. Which molecular defect is most likely?

- A- mRNA splicing mutation in the β -globin gene
- B- Deletion of all alpha-globin alleles
- C- Deletion mutation within the alpha-globin gene
- D- Polyadenylation signal mutation within the alpha-globin gene

33	34	35
D	B	A

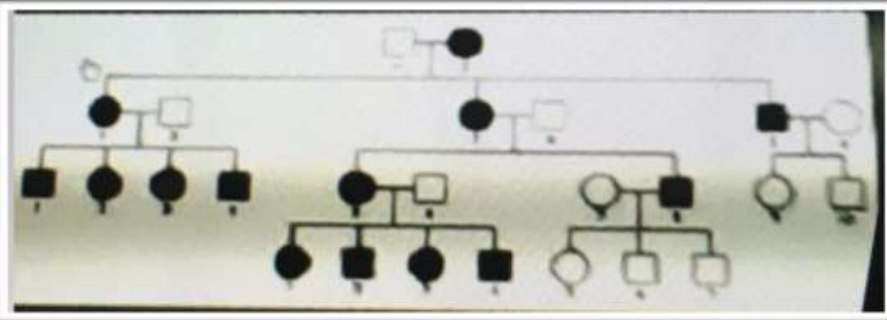
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Genetics Final Med 15

36. Which of the following is most likely a feature of the inheritance pattern shown in the pedigree?

A- Maternal inheritance



For question 36

37. Which enzyme is responsible for repairing pyrimidine dimers (TT or CC dimers)?

- A- Methyltransferase
- B- Low-fidelity polymerase
- C- Ligase
- D- Photolyase

38. Which condition may result from an abnormal mechanical force that distorts a normally developing structure?

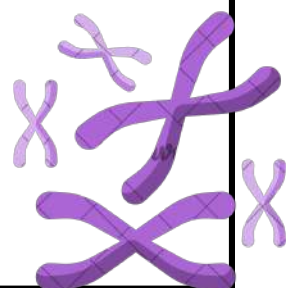
- A- Marfan syndrome
- B- Neural tube defect
- C- Prenatal talipes (clubfoot)
- D- Cleft lip and palate
- E- Achondroplasia

39. The total number of individuals with a given genotype divided by the total population represents:

- A- Heterozygote advantage
- B- Carrier frequency
- C- Phenotype frequency
- D- Mutation frequency
- E- Genotype frequency

36	37	38	39
A	D	C	E

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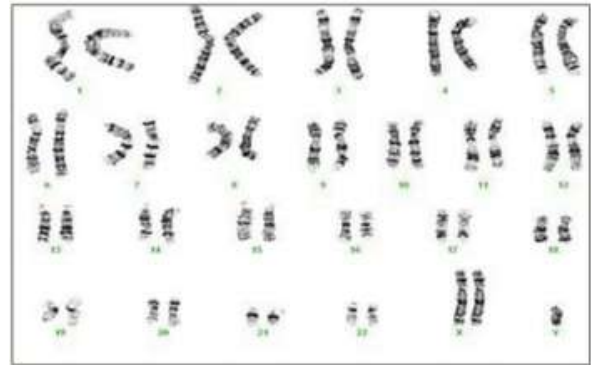
Genetics Final Med 15

40. Low-methionine diet and cysteine supplementation are used in the treatment of:

- A- Hereditary fructose intolerance
- B- Pompe disease
- C- Homocystinuria

41. The following karyotype represents:

- A- Klinefelter syndrome
- B- Patau syndrome
- C- Cri-du-chat syndrome
- D- Turner syndrome
- E- Down syndrome

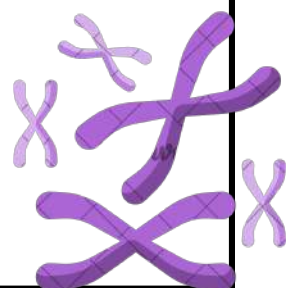


42. Anti-cancer gene therapy includes all of the following EXCEPT:

- A- Insertion of a suicide gene to increase cancer cell death
- B- Insertion of genes that make cancer cells more sensitive to chemotherapy
- C- Prevention of tumor vascularization (anti-angiogenesis)
- D- Insertion of genes that induce vascularization and invasion of cancer

40	41	42
C	A	D

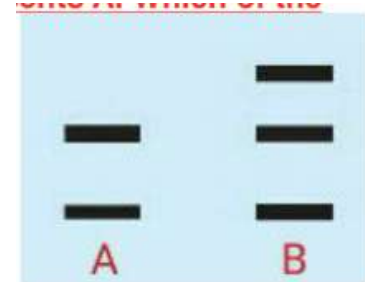
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43. Sarah and Hamzah underwent premarital screening for sickle cell disease. In the RFLP image, Hamzah is lane A and Sarah is lane B. Which statement is correct?

- A- Both have sickle cell trait
- B- Hamzah has sickle cell trait and Sarah is normal
- C- Sarah has sickle cell trait and Hamzah is normal
- D- Sarah has sickle cell disease and Hamzah is normal
- E- Hamzah has sickle cell disease and Sarah is normal



44. The process that results in the formation of the trilaminar germ disc and the three germ layers is:

- A- Gastrulation
- B- Neurulation
- C- Organogenesis

45. What type of mutation does not change the amino acid sequence of the gene product?

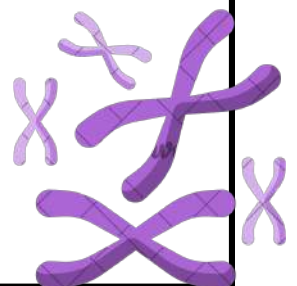
- A- Silent mutation
- B- Missense mutation
- C- Deletion mutation
- D- Non-coding mutation

46. Sickle cell disease results from a single nucleotide substitution that causes a single amino acid substitution. By definition, this is a:

- A- Missense mutation

43	44	45	46
C	A	A	A

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47. High maternal serum alpha-fetoprotein (AFP) levels are most commonly associated with:

- A- Down syndrome
- B- Trisomy 21
- C- Neural tube defect
- D- Trisomy 13

48. A patient develops excessive bleeding while taking warfarin. Which genetic defect is most likely responsible?

- A- CHE mutation
- B- RYR mutation
- C- CYP2C9 mutation
- D- ALDH2 mutation

49. Sulfonamides (sulfa drugs) are contraindicated in individuals with:

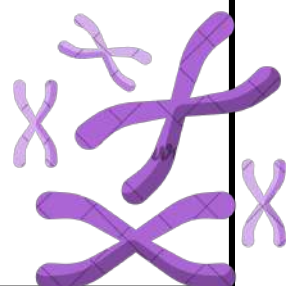
- A- Malignant hyperthermia
- B- G6PD deficiency
- C- Alcohol intolerance
- D- Succinylcholine sensitivity
- E- Debrisoquine metabolism defect

50. Induced pluripotent stem cells (iPSCs) are:

- A- Cells obtained from the inner cell mass of a preimplantation embryo
- B- Stem cells derived from adult cells that cannot differentiate into other cell types
- C- Reprogrammed mature cells that behave like embryonic stem cells without destroying an embryo

47	48	49	50
C	C	B	C

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51. Which of the following syndromes is a classic example of pleiotropy?

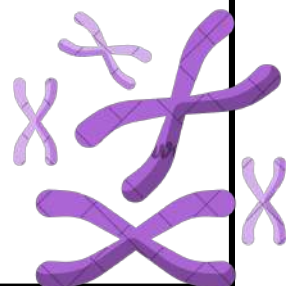
- A- Marfan syndrome
- B- Turner syndrome
- C- Down syndrome

52. The incidence of an autosomal recessive disorder is 1/1000. What is the carrier frequency?

- A- 1/100
- B- 1/50
- C- 1/10
- D- 1/1000
- E- 99/100

51	52
A	B

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CASES

Case 1:

An 8-year-old boy presents with frequent falls and difficulty standing up independently. Physical examination reveals progressive muscle weakness and pseudohypertrophy of the calf muscles. His maternal uncle has similar symptoms.

A. Draw the pedigree

Three-generation pedigree showing X-linked recessive inheritance

B. What is the diagnosis?

Duchenne Muscular Dystrophy (DMD)

C. What is the mode of inheritance?

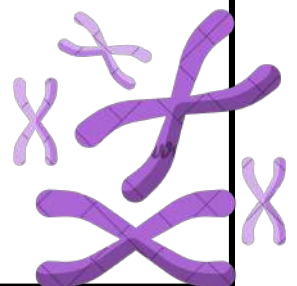
X-linked recessive

D. What test would you use to confirm the diagnosis?

Molecular testing for mutations in the DMD gene (Xp21)

E. What is the recurrence risk?

- 50% affected males
- 50% carrier females



Genetics Final Med 15

CASES

Case 2:

A patient presents with:

- Coarse facial features
- Corneal clouding
- Hepatosplenomegaly
- Skeletal deformities

A. What is found in the urine?

Increased excretion of glycosaminoglycans (GAGs)

B. What is the mode of inheritance?

Autosomal recessive

C. Which enzyme is deficient?

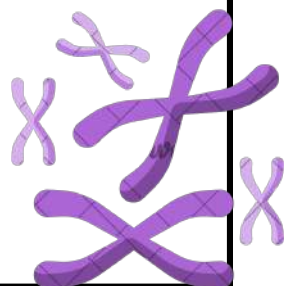
α -L-Iduronidase deficiency

D. What laboratory test confirms the diagnosis?

Molecular genetic testing

E. What is the treatment?

- Enzyme replacement therapy
- Bone marrow transplantation



Genetics Final Med 15

CASES

Case 3:

A. Name the two abnormalities indicated by the arrows:

1. Cleft lip/palate
2. Polydactyly



B. What is the diagnosis?

Patau Syndrome

C. What test would you use to confirm the diagnosis?

Cytogenetic analysis (Karyotyping)

D. What result would you expect?

Trisomy 13

Case 4:

A child presents with recurrent episodes of pain involving the lower back, legs, abdomen, and chest, with periods of remission. The parents are first cousins, and there is a family history of a similar disease.

A. What is the diagnosis?

Sickle Cell Disease (Sickle Cell Anemia)

B. What type of mutation causes the disease?

Point mutation

C. What subtype of mutation is it?

Missense mutation

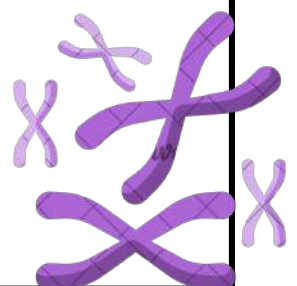
D. What is the mode of inheritance?

Autosomal recessive

E. What is the recurrence risk?

25% per pregnancy

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Genetics Final Med 15

CASES

Case 5

In 1990, the first successful gene therapy was performed on a 4-year-old girl, Ashanti DeSilva, who had Severe Combined Immunodeficiency (SCID) due to ADA deficiency. White blood cells were removed, a normal ADA gene was inserted using a viral vector, and the cells were returned to the patient.

A. What type of gene therapy was used?

Somatic gene therapy

B. What was the route of gene therapy?

Ex vivo gene therapy

C. What type of vector was used?

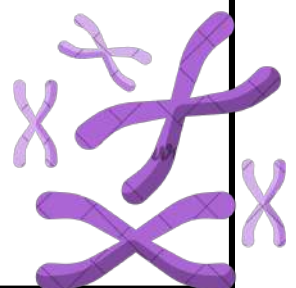
Retroviral vector

D. What are the advantages of this vector?

- Integrates into the host genome
- Long-term gene expression
- High viral titer
- Broad cell tropism
- Relatively straightforward vector production
- Minimal immune response

E. What are the disadvantages of this vector?

- Small insert size (~7 kb)
- Random DNA integration
- Insertional mutagenesis
- Infects mainly dividing cells



Genetics Final Med 17

1. Which of the following is a type of an oncogene?

- A. RB1
- B. BRCA1
- C. TP53
- D. H-RAS

2. In mutational event; when adenine is replaced with guanine it is a case of

- A. Transcription
- B. Transition
- C. Transversion
- D. Frameshift mutation

3. Failure in DNA repair as removing the double-strand break by non-homologous and joining will affect genes such as:

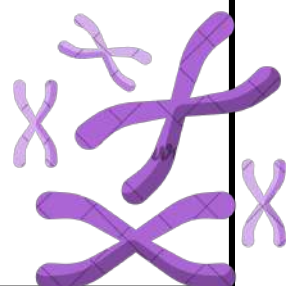
- A. NBS causing bloom syndrome
- B. MYH causing colorectal cancer
- C. XP causing xeroderma pigmentosa
- D. BRCA1/BRCA2 causing breast cancer

4. A neonate admitted to the NICU with ammonia level above 200 $\mu\text{mol/L}$, this will suspect one of the following disorders:

- A. Urea cycle disorder
- B. Mucopolysaccharidosis
- C. Mitochondrial disorder
- D. Glycogen storage disease

1	2	3	4
D	B	D	A

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5. In gene therapy, to be successful, the healthy gene inserted into a target cell must:

- A. Be able to make the correct amount of the protein need
- B. Be inserted manually into the cell mitochondria
- C. Take over and kill the defective gene
- D. Become attached to the cell's mRNA molecules

6. A consanguineous couple came to the genetic clinic seeking for genetic counseling because they have a family history of phenylketonuria. The most favorable counseling should be given to them is:

- A. This is a multifactorial disorder and they have a chance of 4% of getting an affected baby
- B. This is an autosomal recessive disorders, they have 50% chance of being carriers
- C. This is an x-linked recessive disorder and the females have 50% chance of being carriers
- D. This is an autosomal dominant disorder with reduced penetrance

7. A child presented with non-ketotic hypoglycemia, this may be due to:

- A. Von Glerke disease
- B. Hereditary fructose intolerance
- C. Galacosemia
- D. Hyperinsulism

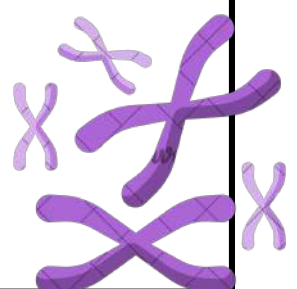
8. Three days old boy came to the ER with a seizure and coma, his blood test showed hyperammonemia

800umol/l. which of the following amino acid changes will be diagnostic for urea cycle disorder?

- A. High Citrulline
- B. High Methionine
- C. High Valine
- D. High Tyrosine

5	6	7	8
A	B	D	A

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9. Which of the following genes is classified as a type of a tumor suppressor gene?

- A. DNA repair gene BRCA2
- B. Growth factors gene – HST
- C. Growth factor receptor gene – RET
- D. Transcription factors gene – MYC

10. A couple came to you after they lost a child with galactosemia and they want to have a healthy baby what would you suggest to them?

- A. Couple carrier screening
- B. Preplantation genetic diagnosis
- C. Prenatal screening by ultrasound
- D. Newborn screening

11. The lab called you to inform you about abnormal results with high serum Leucine. The abnormality is commonly seen in

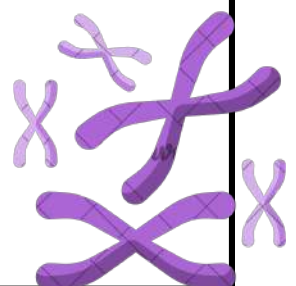
- A. Fructose intolerance
- B. Maple syrup urine disease
- C. Urea cycle defect
- D. Homocystineuria

12. Which gene transfer technique involves the use of fatty bubble to carry a gene into a somatic cell?

- A. Microinjection
- B. Liposomes
- C. Particle bombardment
- D. Electroporation

9	10	11	12
A	B	B	B

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13.Unacceptable (prohibited) option in genetic counseling in our society is:

- A. Artificial insemination using donor sperm
- B. Premarital genetic diagnosis
- C. Neonatal screening and treatment
- D. Premarital carrier detection

14.Which of the following prenatal test doesn't carry a risk of fetal loss?

- A. Maternal serum AFP
- B. Chronic villus sampling
- C. Fetal tissue sampling
- D. Amniocentesis

15.What is the unique thing about stem cells that makes them so desirable for therapeutic use?

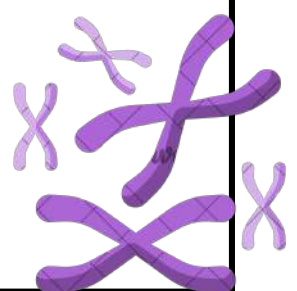
- A. They develop into nerve cells and can be used by stroke victims
- B. They are found in reproductive tissue
- C. They can develop into many different tissue types
- D. They develop into organs needed for transplant

16-what are induced pluripotent stem cells (iPS cells)?

- A. A genetically reprogrammed mature cell that acts just like an embryonic stem cell but without the need to destroy an embryo
- B. Cells found in the inner mass of blastocyst that can differentiate into over 200 different cell types except placental cells
- C. Stem cells derived from adult tissue and do not differentiate into any other cell type
- D. Stem cells derived from adult tissue, cord blood, and bone marrow and that do not differentiate into all the different cell types

13	14	15	16
A	A	C	A

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17. Retroviral vectors are effective means for gene therapy as they:

- A. Can readily affect non-dividing cells
- B. Remain extra chromosomal
- C. Stably integrate into the host cells genome
- D. Stimulate host immune response to the infected cells

18. Post subject: genetically modified Retrovirus saves SCID patient: on September 14 1990, the first gene therapy was performed on a 4-year-old girl, Ashanti De Silva who Was treated for severe combined immunodeficiency (SCID). Doctors collected samples of Ashanti's blood, isolated some of her white blood cells, and used a virus to insert a healthy adenosine deaminase (ADA) gene into them. The type of gene therapy applied to this case is:

- A. Stem cell gene therapy/Ex vivo gene therapy
- B. In vivo gene therapy/
- C. Germ line therapy

19. Individuals with sickle cell trait:

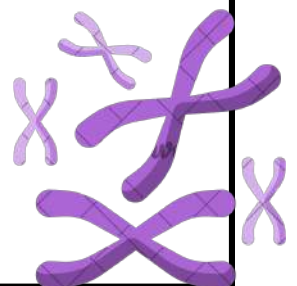
- A. Carriers of an X-linked recessive trait
- B. Homozygous for the mutation
- C. Heterozygotes, therefore relatively protected from malarial attacks
- D. HbF production persists into childhood and beyond

20. Concerning Cooley's anemia:

- A. There is underproduction of alpha globin chain
- B. Usually present in infancy with a severe transfusion dependent anemia
- C. There is substitution of valine for glutamic acid in the B chain
- D. Analysis of hemoglobin from such patients reveals a tetramer of gamma globin chains

17	18	19	20
C	A	C	B

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21. Two first cousin couple whom are still planning to marry are worried about having kids with Phenylketonuria as they have a family history of the disease in their relatives. The best intervention you can provide them:

- A. Preimplantation genetic diagnosis
- B. Premarital carrier screening
- C. Prenatal screening by ultrasound
- D. Newborn screening

22. Glucose 6 phosphate dehydrogenase deficiency is responsible for one of the following scenarios:

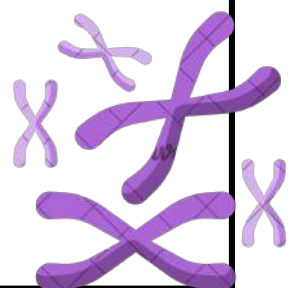
- A. A girl 12-year-old, complained of vertigo and numbness in hands after treatment with isoniazid
- B. A male 62-year-old, treated from a stroke, develop bleeding after using an anticoagulant
- C. A boy 8-year-old, developed acute hemolysis after ingestion of sulpha now preparation
- D. A female 24-year-old, developed high grade fever during surgical intervention

23. All cells contain the same genetic information. Why can't cells other than stem cells differentiate into various tissues?

- A. Stem cells are the only cells that do not have an X or Y chromosome and therefore go into either a male or female
- B. As cells develop their genetic makeup change
- C. As cells develop some genes are turned off permanently
- D. Stem cells have haploid chromosome number

21	22	23
B	C	C

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24. Direct DNA injection of naked DNA plasmid is currently used in:

- A. Treatment of cystic fibrosis
- B. Long term expression of transgene
- C. High level of expression of transgene
- D. Vaccine development

Answer is: D

25. The commonly used biochemical markers and triple screen test in maternal serum included all the following except:

- A. Maternal serum AFP
- B. Indirect serum bilirubin
- C. hCG
- D. Unconjugated estriol

26. A mutation in the ryanodine (RYR1) gene is responsible for:

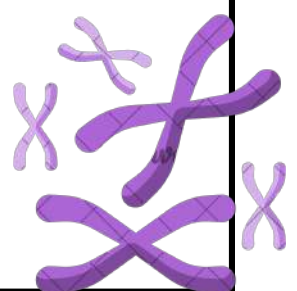
- A. Malignant hyperthermia when halothane is used as the anesthetic with succinylcholine
- B. Debrisoquine toxicity
- C. Rapid metabolism of isoniazid
- D. Decreased metabolism of warfarin

27. The first gene therapy clinical trial for treatment of Ornithine Transcarbamylase deficiency used a viral vector to introduce the normal OTC gene directly into the patient's liver. He died within hours due to immune response. Based on this information select the correct sequence of answers to the following questions; What type of gene therapy has been applied? What is the mode of gene therapy used? What type of vector has been used?

- A. Germ line—ex vivo—Adenovirus
- B. Germ line—in vivo—Retrovirus
- C. Somatic—ex vivo—Retrovirus
- D. Somatic—in vivo—Adenovirus

24	25	26	27
D	B	A	D

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28. Inhibin-A Marker used to show an increased level in maternal serum of:

- A. Turner syndrome
- B. Down syndrome pregnancies
- C. Cystic fibrosis pregnancies
- D. Duchenne muscular dystrophy pregnancies

29. A three-year-old boy was being treated for SCID using a modified Retrovirus, but the boy developed a leukemia like condition. What appeared to have caused this?

- A. The Retrovirus inserted near a proto-oncogene promoter
- B. The disorder is known to cause leukemia in most patients
- C. The Retrovirus triggered the human T cell leukemia virus
- D. The viral batch was contaminated with RCRs

30. Once a therapeutic gene has been integrated into a target cell's DNA, the Retroviral DNA will most likely:

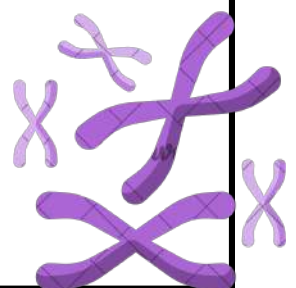
- A. Replicate and form infectious virions
- B. Persist in the cell in a non-infectious form
- C. Be recognized as "foreign" by the host's immune system and degraded
- D. Cause a nondisjunction event that corrects the genetic defect

31. When describing the Knudson "two hit" hypotheses regarding sporadic tumors it requires tumor suppressor gene to be mutated by:

- A. 1 acquired mutation, 1 hit on somatic level
- B. 1 inherited mutation
- C. 2 acquired mutations, 2 hits on somatic level
- D. 3 acquired mutations, 1 hit on somatic level

28	29	30	31
B	A	B	C

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32. Pseudocholinesterase deficiency is associated with increased sensitivity to what?

- A. Primaquine
- B. Halothane
- C. Succinylcholine
- D. Fava beans

33. A female patient six-year-old presented with intellectual disability, seizures, fair color of hair and skin, as well as a musty odor of sweat and urine. In this disorder:

- A. There is a deficiency of enzyme required for the conversion of Phenylalanine to tyrosine
- B. Dietary restriction of fructose is associated with a good long term prognosis
- C. Treatment involves a low-methionine diet with cysteine supplementation
- D. The diagnosis confirmed by demonstration of the branched chain amino acids in blood

34. Reverse transcriptase polymerase chain reaction (RT-PCR) is a method that is used for:

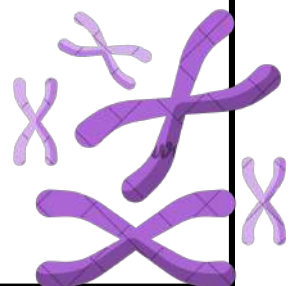
- A. Amplification of mRNA sequences
- B. Amplification of amino acid sequences
- C. Forensic analysis of DNA
- D. Amplification of genomic DNA sequences

35. Retrovirus uses which enzyme to convert information stored in their RNA to DNA?

- A. RNA polymerase
- B. DNA ligase
- C. Reverse transcriptase
- D. A restriction enzyme

32	33	34	35
C	A	A	C

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36. Structural globin chain variants include:

- A. Hereditary Persistence of Fetal Hemoglobin
- B. Mediterranean Sea anemia
- C. Sickle cell disease
- D. Hydrops Fetalis

37. Germline gene therapy could permanently correct a genetic defect in a(n):

- A. Parent of an affected child
- B. Affected individual and all of his or her descendants
- C. Affected individual and all of his or her offspring only
- D. Affected individual only

38. A male baby, four-month old presented with jaundice since birth, cataract and hepatomegaly investigations revealed a reducing substance in the urine. The management protocol of such disorder should be:

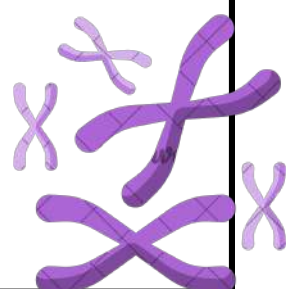
- A. Restriction of fructose in diet
- B. Restriction of Leucine and Valine from diet
- C. Giving tyrosine supplements
- D. Prescribing a lactose free milk formula

39. A 5-year-old girl was evaluated in the clinic as a case of intellectual disability and myopia, laboratory evaluation revealed elevated serum methionine. which of the following is the cause for intellectual disability?

- A. High ammonia when you live the leading to arginine deficiency
- B. High homocysteine level leading toxic injury to the brain
- C. High galactose level leading to lack of injury in the brain
- D. High cystathionin level leading to toxic injury to the brain

36	37	38	39
C	B	D	B

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Genetics Final Med 17

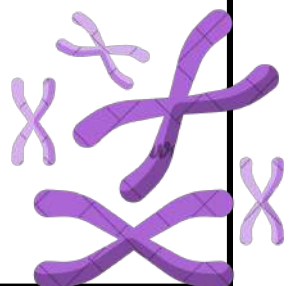
40. Which of the following is preferred vector for the treatment of neurological disorders such as Parkinson's disease?

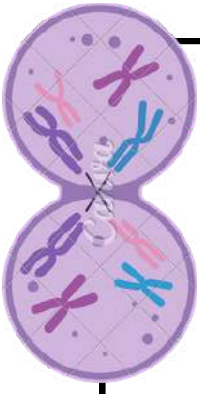
- A. Herpes simplex viral vector
- B. Adenovirus
- C. Adeno-associated viral vector
- D. Cationic liposomes

40

A

في حال وجود أي خطأ أو تعديل الرجاء إرسال
#تعديل_جينيتكس + رقم الصفحة والسؤال والإجابة
الصحيحة إلى الجروب الآتي : {اضغط هنا}





الحمد لله الذي هدانا لهذا وما كُنَّا لِنَهْتَدِي
لولا أن هدانا الله

